

Industrial Internship Project Presentation

Athena

Realtime Recommender System

Prashant Sinha

05-07-2017

ORGANISATION

Agora NHS

London, UK

EXTERNAL MENTOR

Dr. Leif Denby

University of Leeds

INTERNAL MENTOR

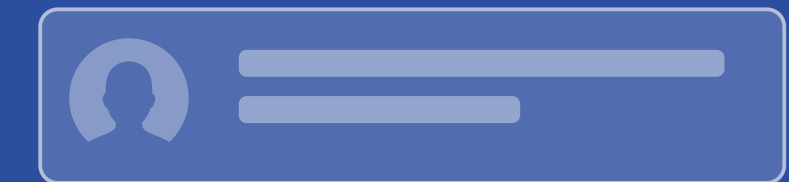
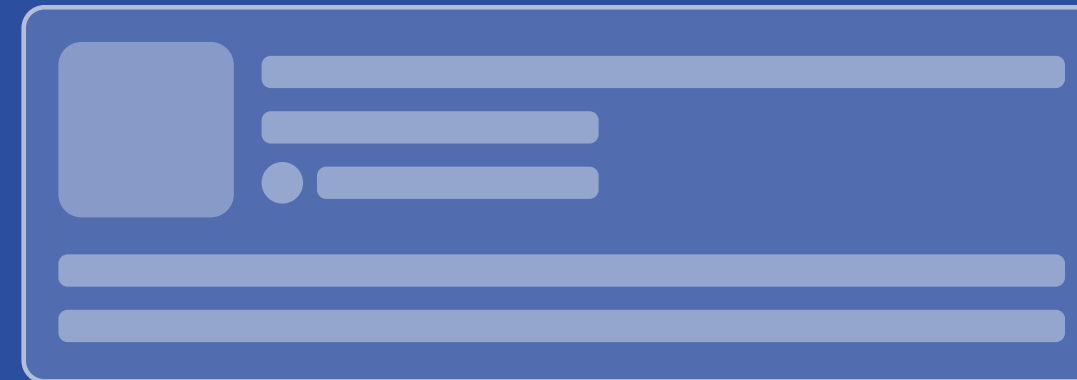
Dr. Sonam Singh

Cluster Innovation Centre, University of Delhi

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#Background

- We have a set of Events, and a set of Users.
- Given a *user*, we want to choose all the *events* that are relevant to that *user*.



#Agenda

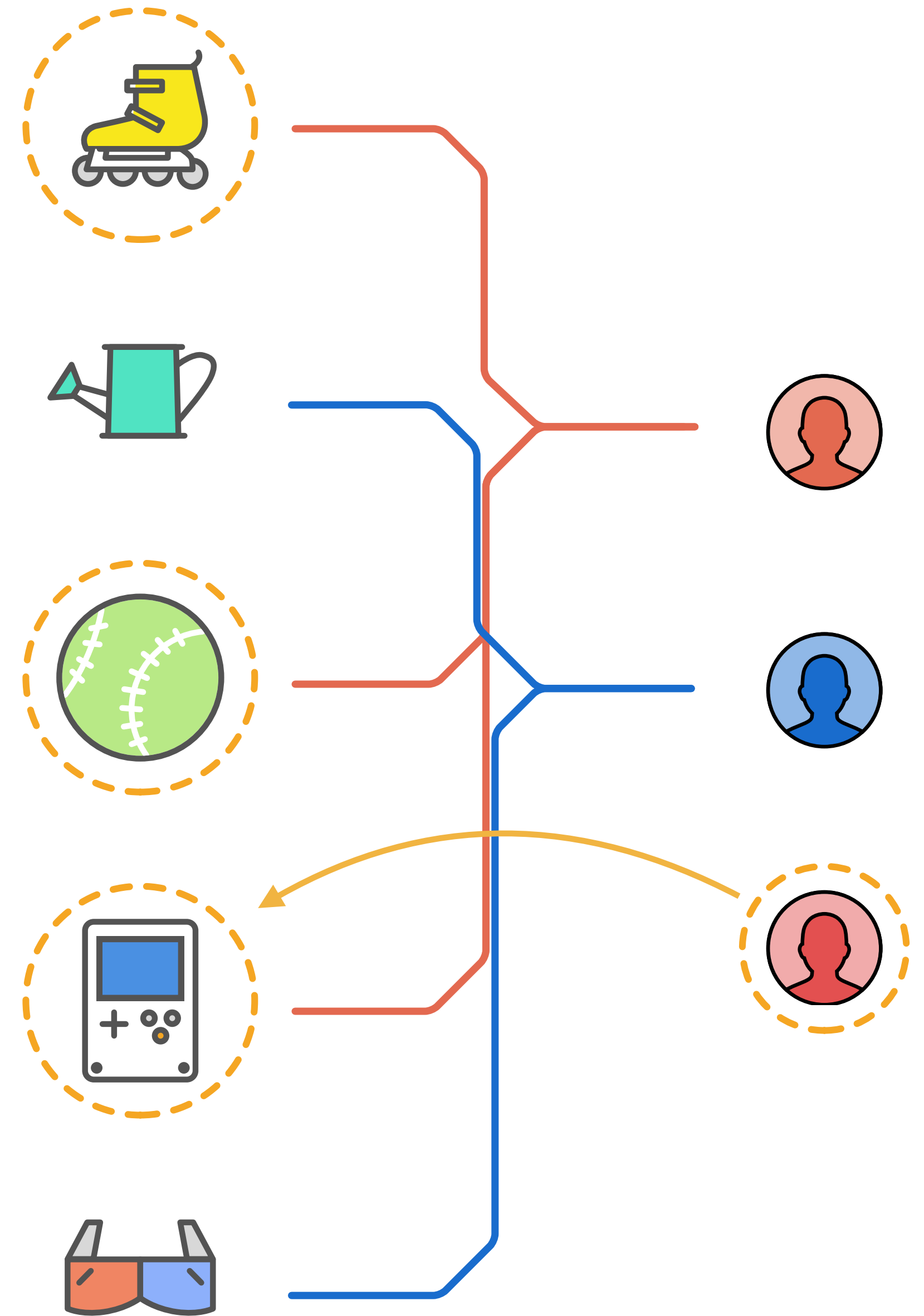
- Recommender Systems
- Performance at Scale
- Elasticsearch Extension
- Outcomes

#Recommender Systems

You might be interested in...

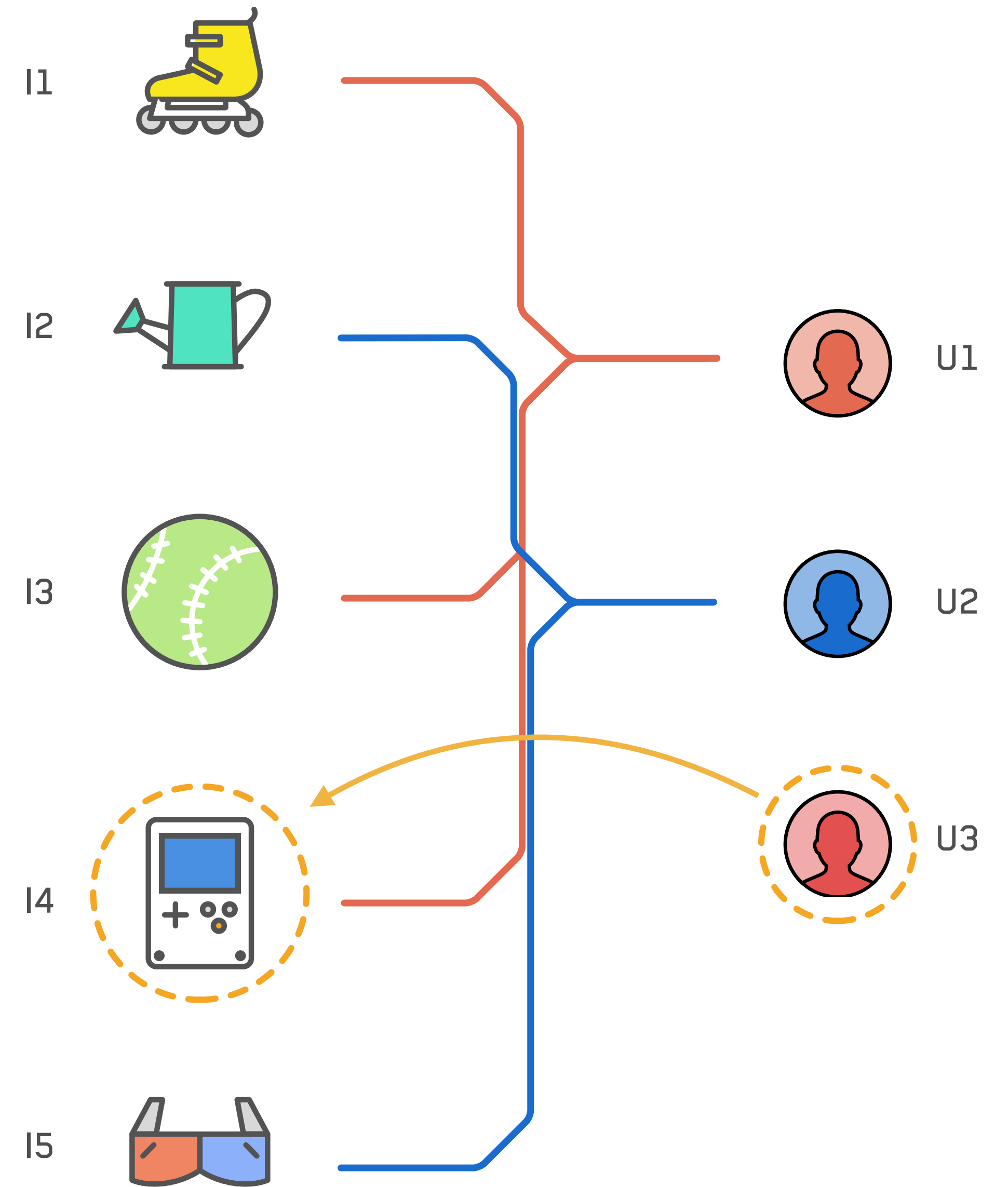
Collaborative Filtering

Find and match *similar users*.



Collaborative Filtering

	I1	I2	I3	I4	I5
U1	1	0	1	1	0
U2	0	1	0	0	1
U3	?	?	?	1	?



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Applied Cryptography

by **Bruce Schneier** (Author)

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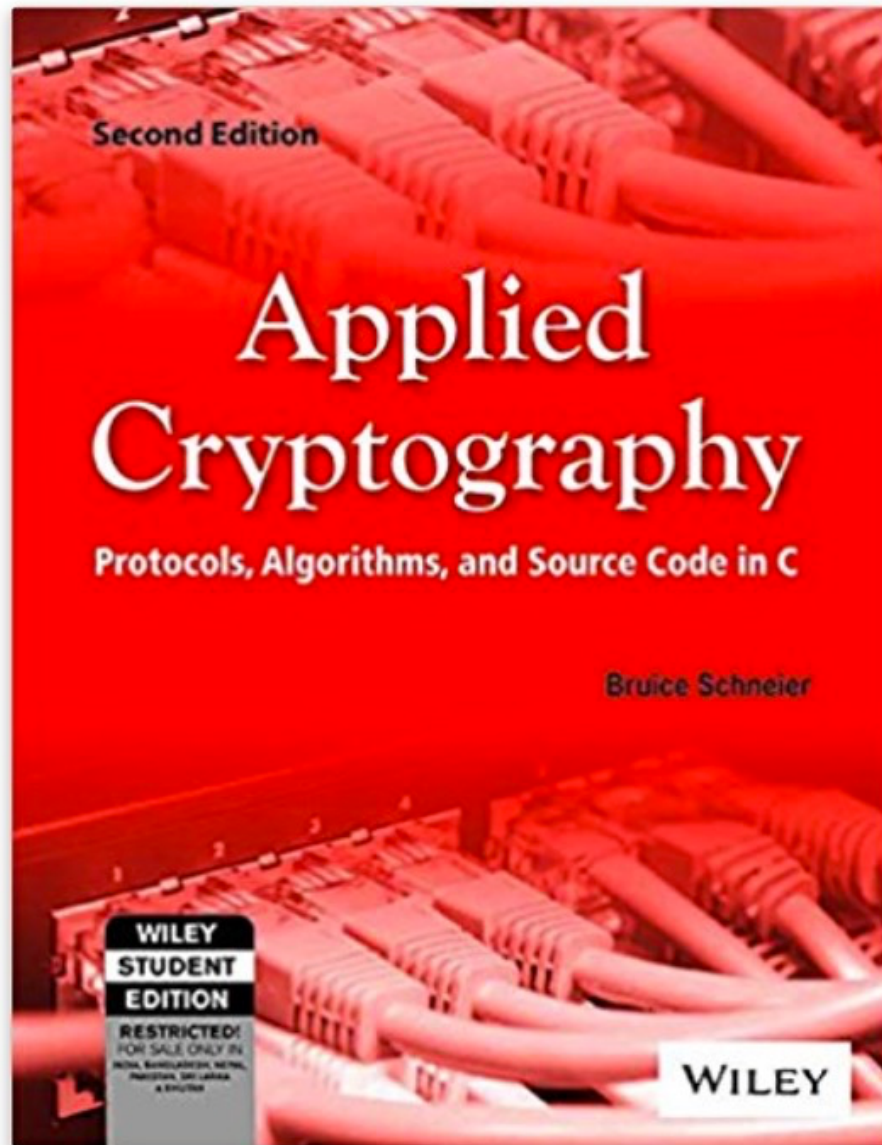
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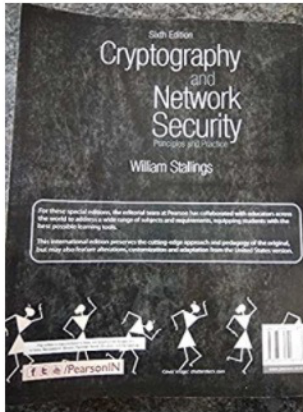
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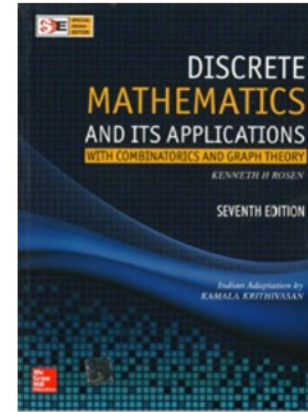
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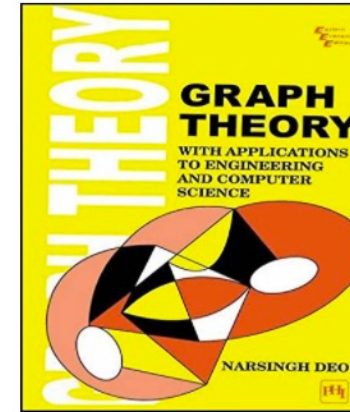
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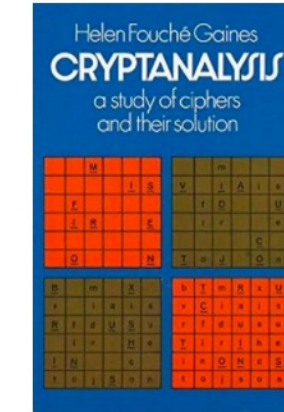
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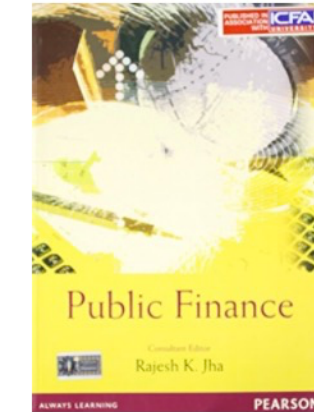
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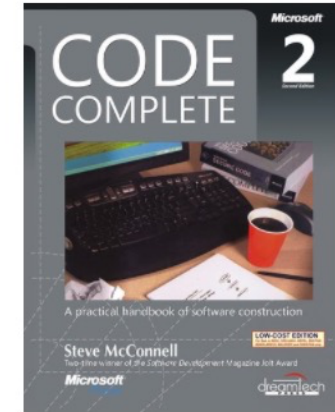
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Simon Singh
★★★★☆ 11
Paperback
₹ 343.00 ✓prime



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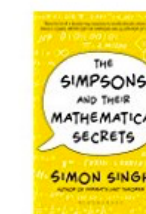
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The Code Boo: The Secret History of Codes and Code - Breaking Paperback
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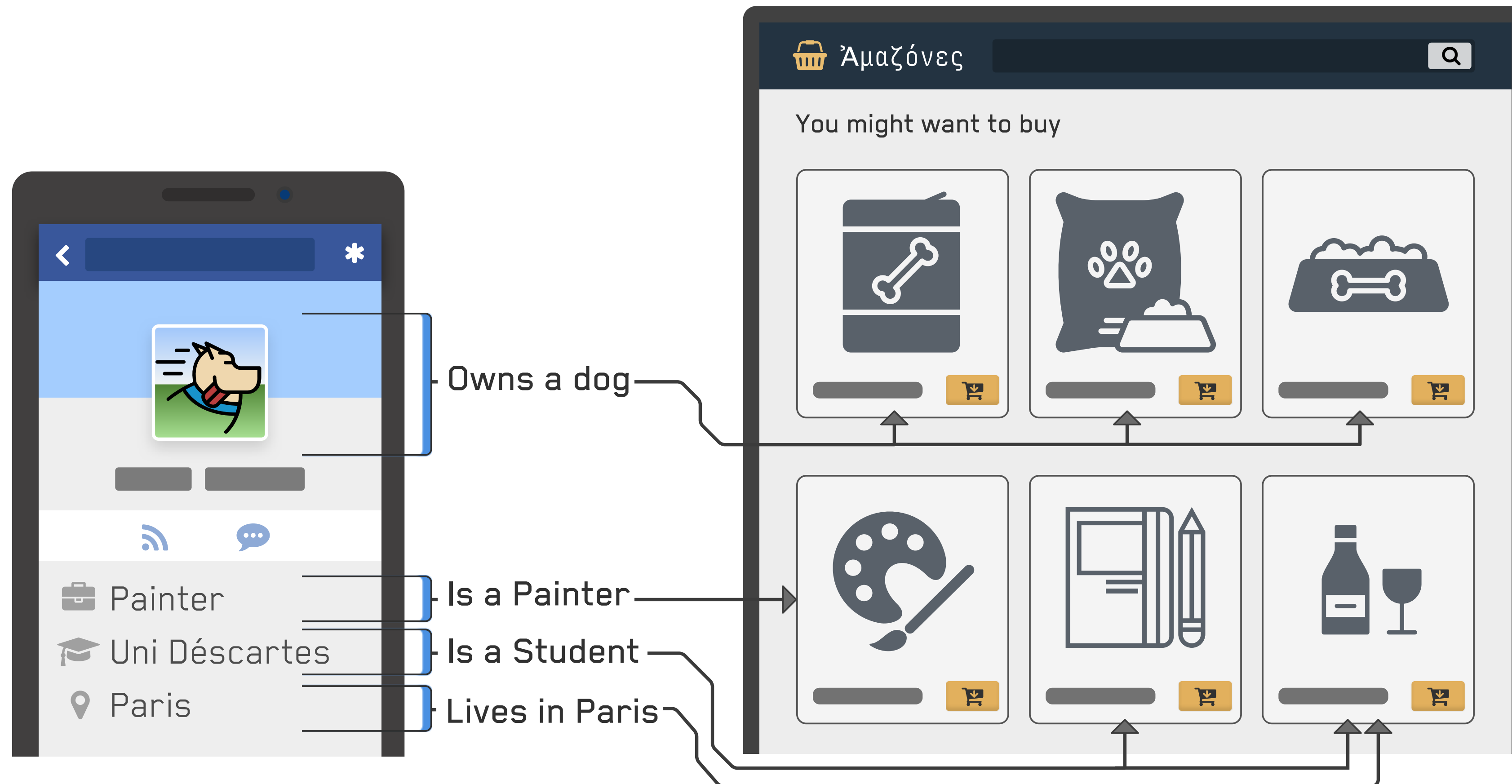
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#Recommender Systems

Content Based



NETFLIX

Browse ▾

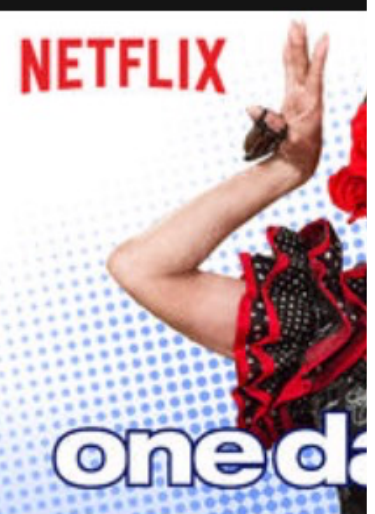
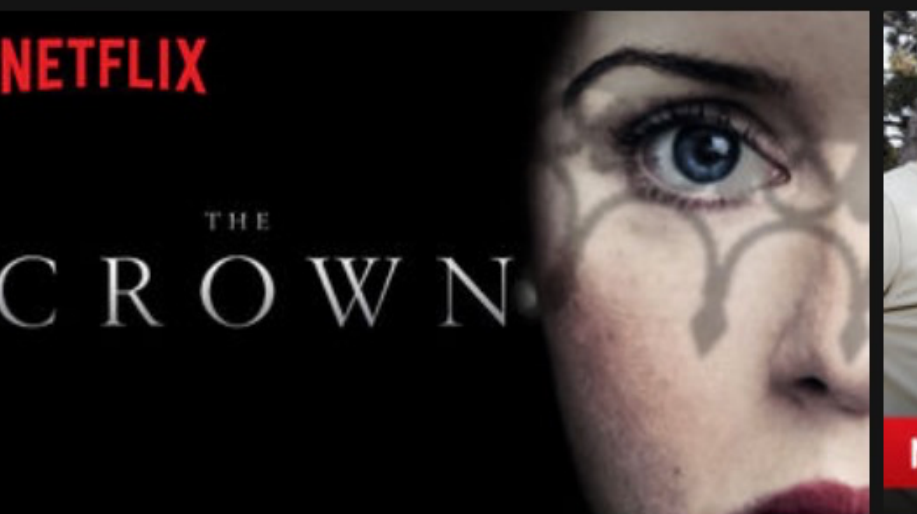
Trending Now



Because you watched Marvel's Luke Cage

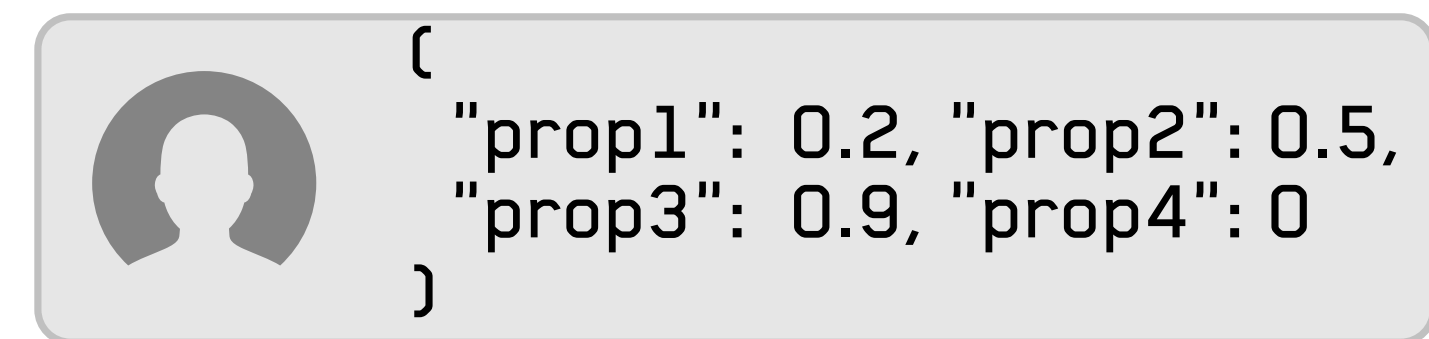
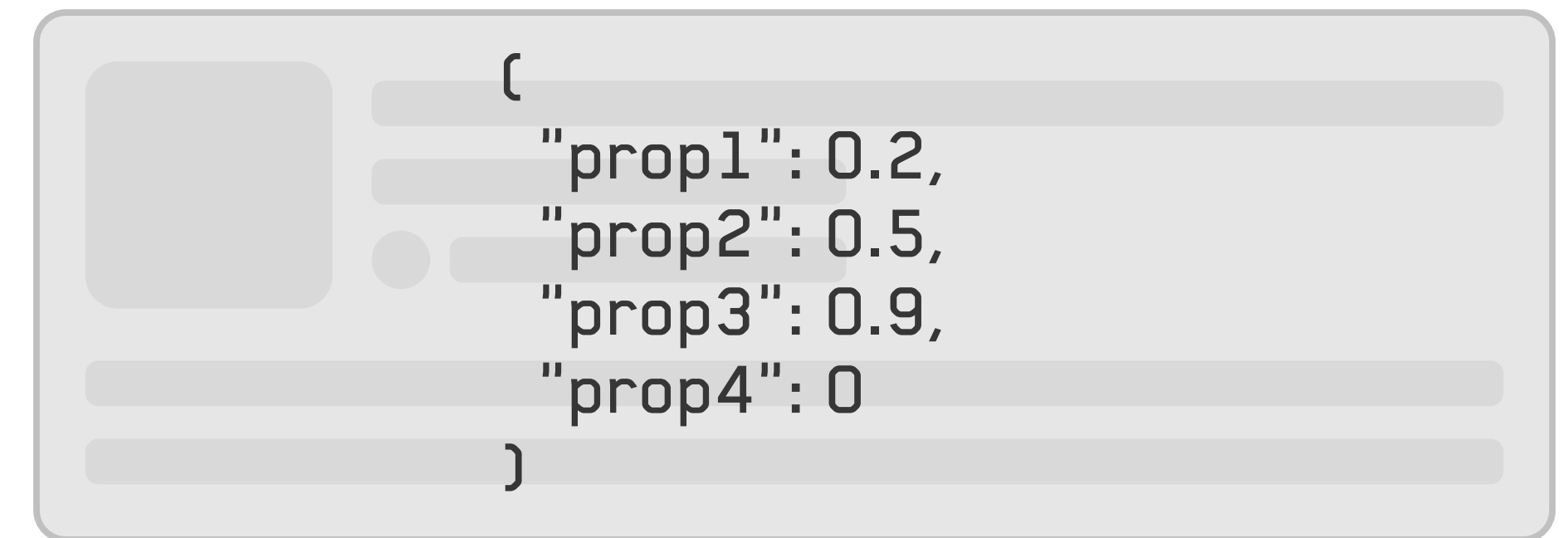


International TV Shows



Content Based

Build *vector-space*
representation of both *event*
and *user*.



#Recommender Systems

Content Based

Relevance score is the cosine product of the vectors.

$$\text{score}(d_u, d_i) = \vec{v}_{d_u} \cdot \vec{v}_{d_i}$$



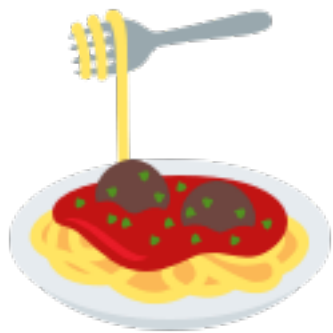
#Recommender Systems

Attribute Discovery



#Recommender Systems

Attribute Discovery



"Happiness"

#Recommender Systems

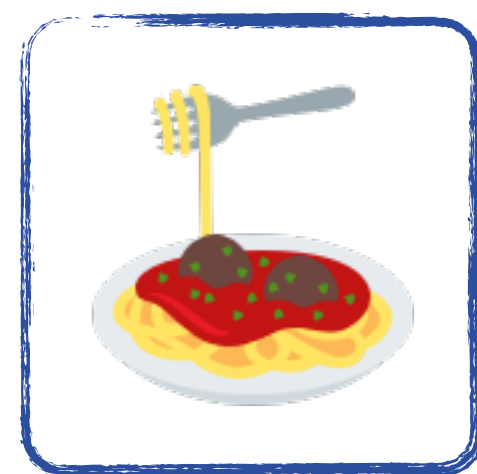
Attribute Discovery



"Distress"

#Recommender Systems

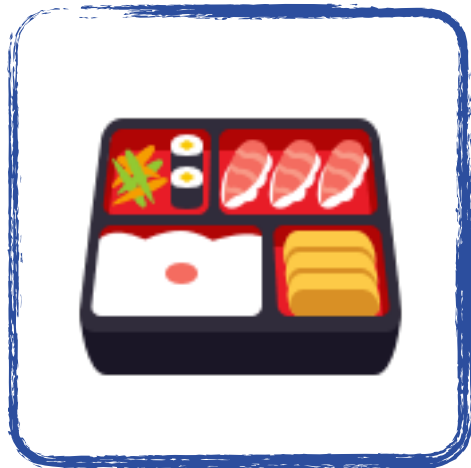
Attribute Discovery



"Edibles"

#Recommender Systems

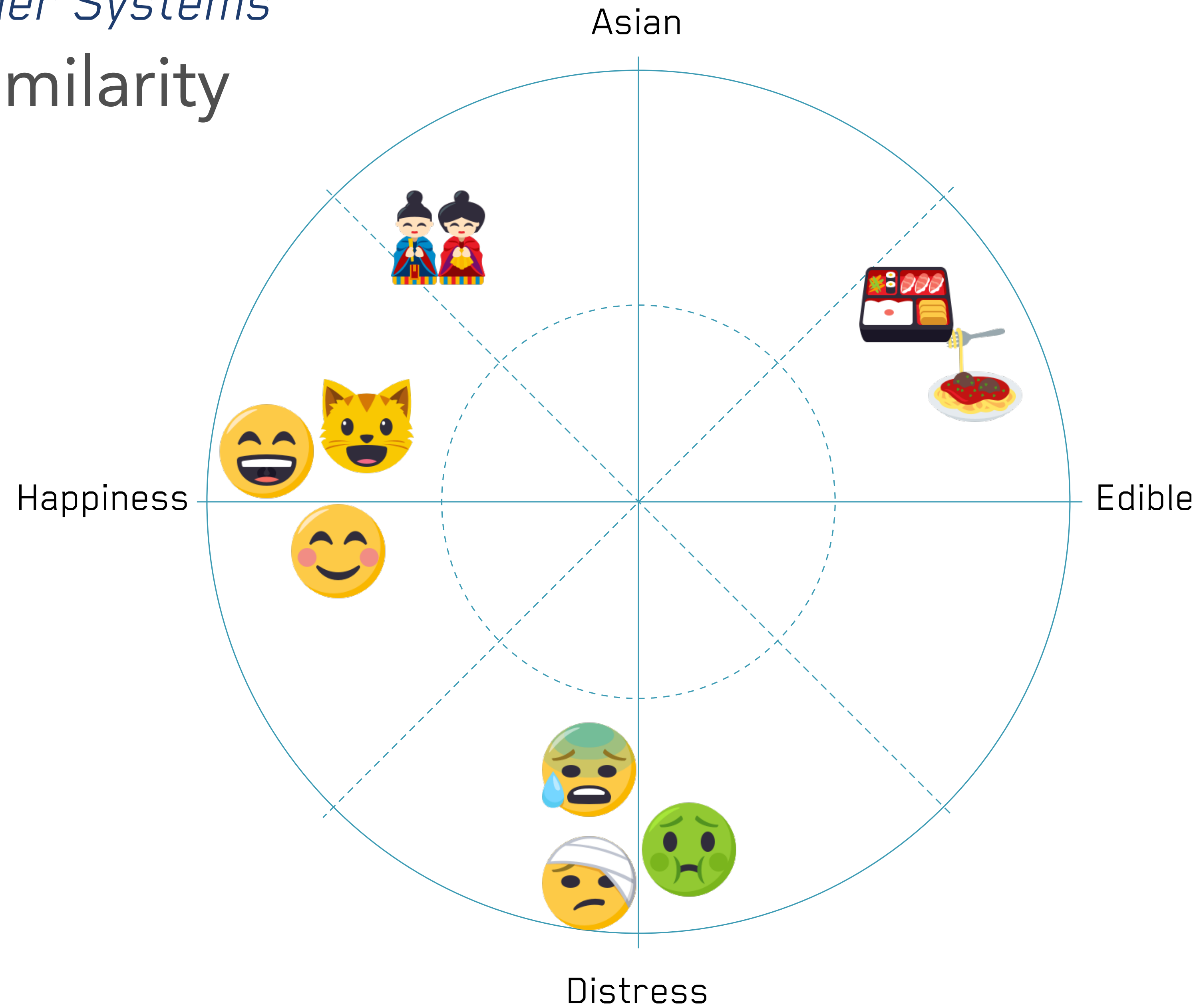
Attribute Discovery



"Asian"

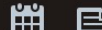
#Recommender Systems

Vector Similarity



#Attribute Discovery in Athena

...



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Tell us about you

1

First Name

Last Name

Prashant

Sinha

What hospital/institute do you work at?

Evelina London Children's Hospital

x

▼

What is your current profession?

Dietetics

x

▼

What is your current training level?

Apprentice

x

▼

What is your current training band?

Band 2

x

▼

What is your professional specialty?

x

Paediatrics

x

Neonatology

x

▼

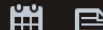
Do you have any additional email?

someone@example.com

Next

>

...



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What other specialties are you interested in?

2



Paediatrics



Neonatology



Anaesthesiology



Cardiology



Clinical Immunology



Dental



Dermatology



ENT



Emergency Medicine



Endocrinology



Epidemiology



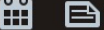
Forensic Medicine







...



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What types of events are you interested in?

3



Clinical governance



CPD Courses



Exam Revision



Introductory courses



Lectures



Non-clinical events



Other/ Not Applicable



Patient outreach



Research Conferences



Seminars



Social events



Workshops

< Previous

Next >

#Attribute Discovery in Athena

...



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Create Event | Details

1

Event title

Event date and time





[Provide end date and time?](#)

Who's the organiser?



What's the location?



Additional Information

Select event's visibility

...



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Create Event | About

2

Which medical fields is your event relevant to?

x

Anaesthesiology

x

Urology

x

Cardiology

x

Description

B

I

U









Normal







Es ist eine bekannte tatsache, dass Hunde sind suss.

Oder ist es?

Wir werden dies in der Veranstaltung diskutieren.

Which training levels is it relevant to? OPTIONAL

x

Apprentice

x

SAS Doctor

x

What type of event is it? OPTIONAL

x

Non-clinical events

x

Research Conferences

x

<

Previous

Next

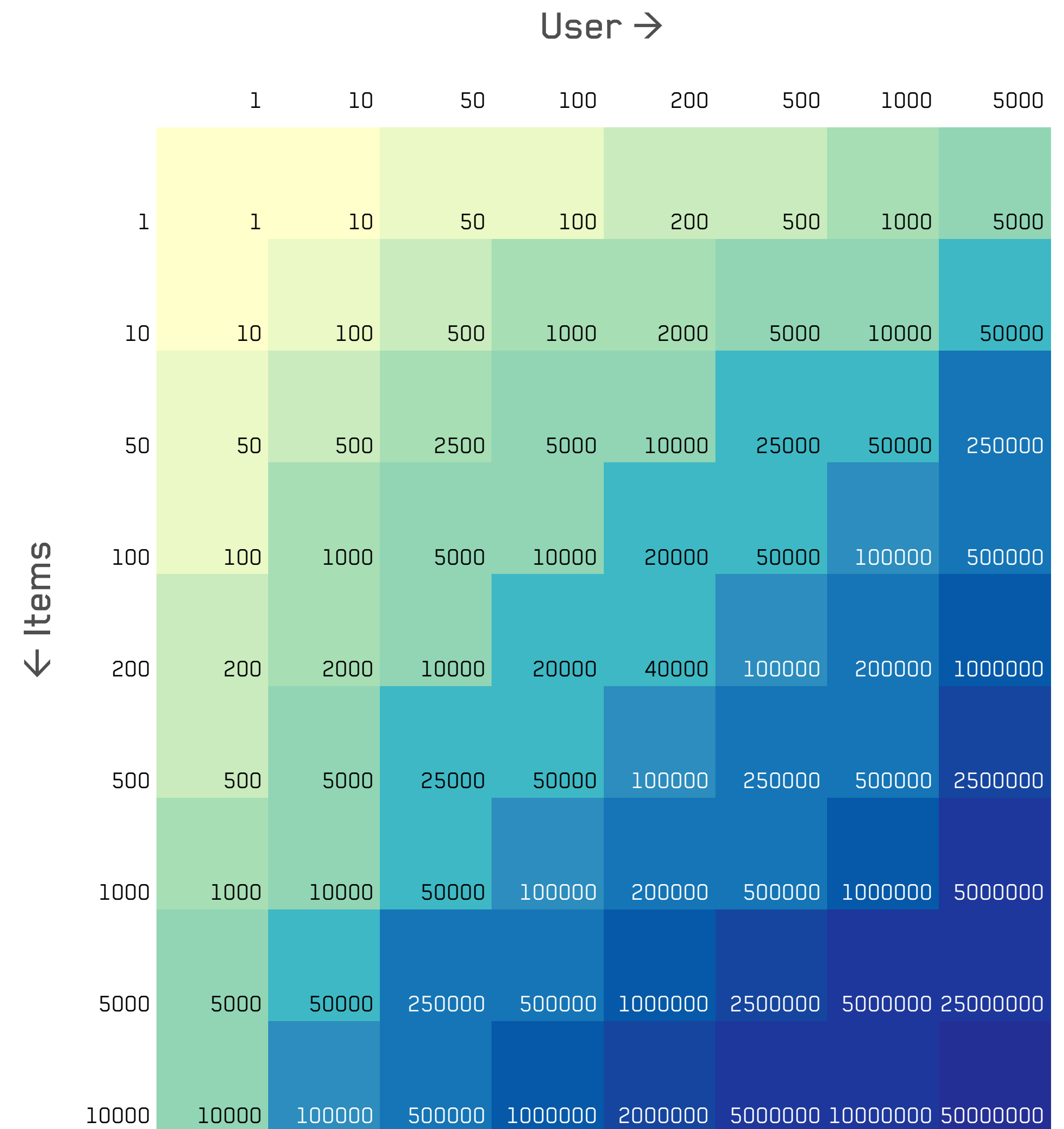
>

#Performance

Performance at Scale

#Performance at Scale

Number of calculations grow rapidly as either of the set grows.



#Performance at Scale

Strategies

Calculate relevancy score in advance.

#Performance at Scale

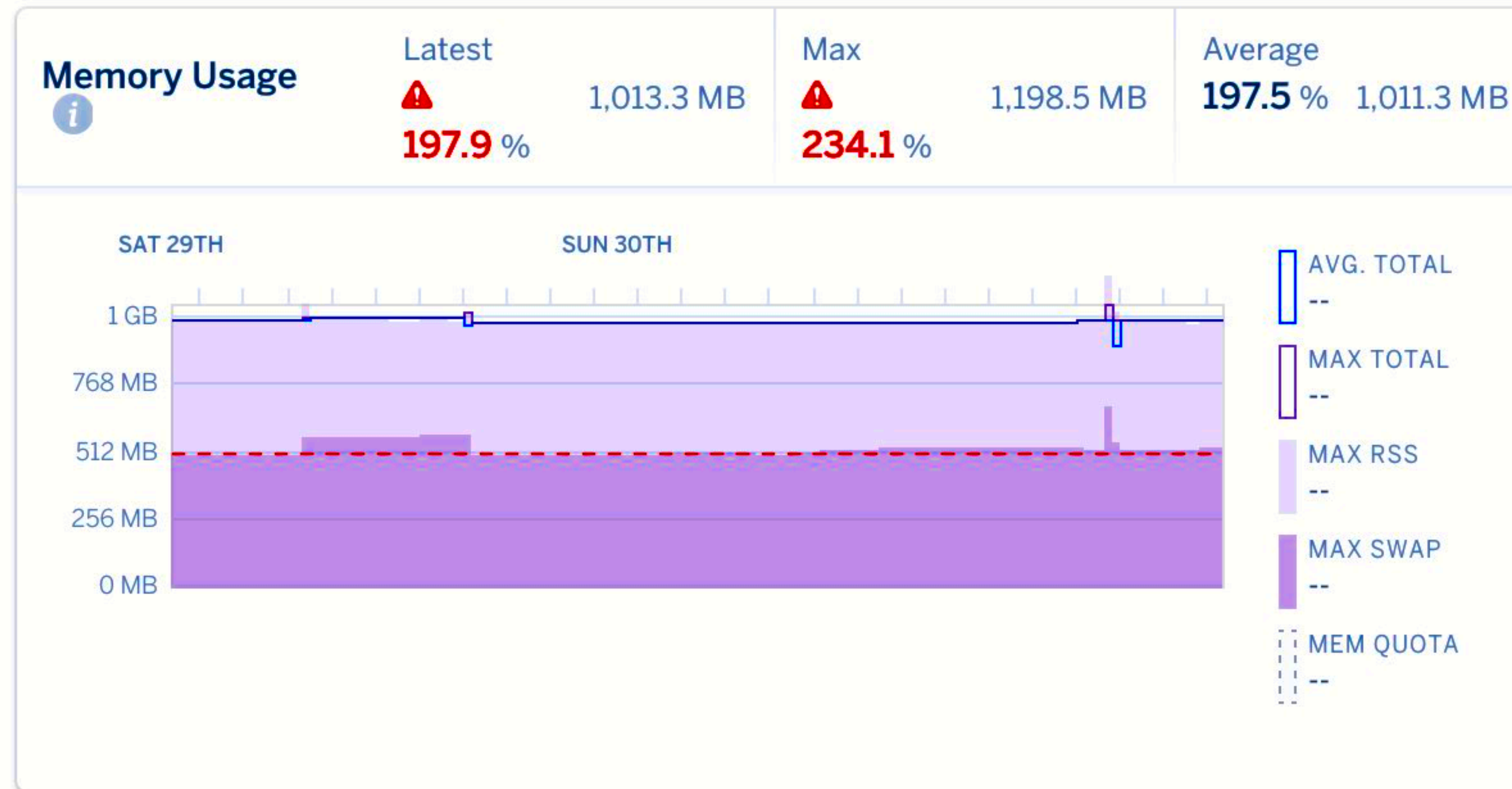
Strategies: Pre-calculating Scores

How does the system perform when...

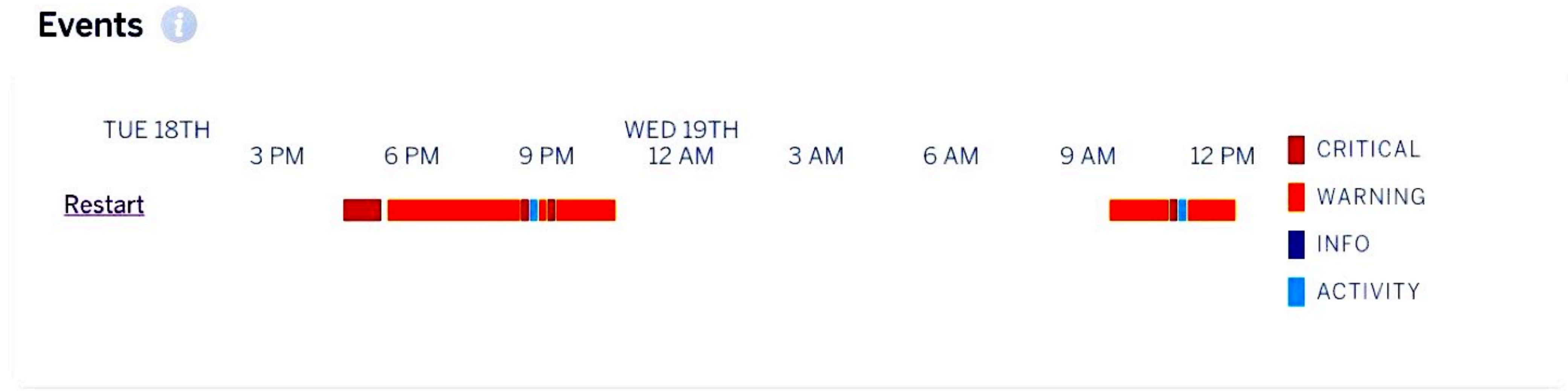
- user profile updates?
- new events are added?

#Performance at Scale

Strategies: Pre-calculating Scores



Strategies: Pre-calculating Scores



Strategies: Pre-calculating Scores

How does the system perform when...

- user profile updates?
- new events are added?

Poorly.

#Elasticsearch

Realtime Scoring using Elasticsearch

#Elasticsearch

Elasticsearch is a distributed search and analytics engine.

#Elasticsearch

Realtime scoring is expensive.

#Elasticsearch

Identify most expensive step.

#Elasticsearch

Perform expensive operation in advance.

During query, perform the less expensive cosine similarity operation.

#Athena

Outcomes

#Elasticsearch



#Outcomes

...

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Showing events for Specialty this week next week week after next all

My Trust

My Specialty

All Events

My Events

28th Annual Meeting of the European Socie...

6th Sep

Recommend

Calendar

REaSoN 2017

2nd Jul

Neonatology
Warwick Arts Centre, University of Warwick,

Recommend

Calendar

6th Annual Neonatal Simulation Conference

26th Sep

Recommend

Calendar

Neonatal Cardiology & Haemodynamics C...

10th Oct

Recommend

Calendar

Dartford Flexible Cystoscopy Course

17th Nov

An interactive course designed for nurses
and doctors with experience of, or intending

Recommend

Calendar

Neonatology

Paediatrics

Dermatology

#Outcomes

...



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Showing events for Specialty this week next week week after next all



Women's Mental Health Research

4th Jul

Women's Mental Health Research
This three day course aims to develop

 Recommend

 Calendar



1st Annual UK Maternal Mental Health Alli...

13th Sep

Join national and international experts in
perinatal mental health in discussing the

 Recommend

 Calendar



Hepatobiliary and pancreatic cancer: What...

20th Sep

The programme for this day meeting will be
available soon. To be notified when

 Recommend

 Calendar



4th Breast cancer forum: screening, metas...

15th Sep

We are delighted to partner with Pink
Ribbon to deliver the fourth breast cancer

 Recommend

 Calendar



H3 symposium: Sensory transduction in ins...

8th Dec

More information will follow.

 Recommend

 Calendar



The 17th International Cancer Imaging Soci

and Oct



Oncology



Psychiatry



Psychology

37

#Outcomes

Open Source Components

`github.com/prashnts/dj-elasticsearch-flex`

`github.com/prashnts/elasticsearch-synonyms`

#Questions?